

Bishal Marasini

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Research Interests

- Aircraft design and analysis
- Flight dynamics and control
- Aerodynamics and computational fluid dynamics
- Unmanned aerial vehicle design and autonomous systems
- Aircraft performance and flight simulation

Education

Bachelor's Degree in Aerospace Engineering 2022–2026

- Pulchowk Campus, Institute of Engineering, Tribhuvan University, Nepal
- Percentage: 77.42%

Relevant Coursework: Fluid Dynamics, CAD & Manufacturing, Control System, Aerodynamics, Strength of material, Aircraft Propulsion, Flight Dynamics, UAV Synthesis, FEM, Aircraft Preliminary Design, CFD, Aircraft Structure, Fundamentals of International Aviation, Fleet Planning. Thesis: "Crop Health Monitoring Through Aerial Mapping Using Quadcopter-Based Imaging."

Professional Experience

Trainee Performance Engineer (OJT), Operations Department, Yeti Airlines Sept 2025- Mar 2026

Tilganga, Kathmandu

- Performed flight-performance calculations for takeoff, landing, and climb under varying aircraft weight and environmental conditions
- Prepared and verified weight and balance, load and trim sheets, and last-minute changes for daily flight operations
- Assisted with flight planning, fuel estimation, and payload optimization in accordance with aircraft flight-manual and operational limits
- Analyzed runway and high-altitude aircraft performance while supporting operations-control-center dispatch coordination

Research and Design Projects

Crop Health Monitoring and Localization Through Aerial Mapping Using Quadcopter-Based Imaging [THESIS] 2025-2026

- Developed a quadcopter-based aerial mapping workflow for crop health monitoring and localization of affected regions
- Applied computer vision and Random Forest machine learning for crop-health classification and georeferenced mapping
- **Tools:** Pix4D, Python, Drone Deploy, QGIS

Design and Analysis of Sky-Yak: 19-Seat STOL Aircraft 2025

- Performed aircraft sizing, performance, stability, and dynamic analysis for a 19-seat STOL aircraft
- Conducted flight tests using an aircraft simulator to evaluate V-n diagram, CG travel, stall, takeoff and landing characteristics, and handling qualities
- **Tools:** XFLR5, Plane Maker, X-Plane, MATLAB

Design and Analysis of Hawk-Eye: Wildlife Surveillance UAV 2024

- Performed UAV sizing, performance, stability, and dynamic analysis for a wildlife-surveillance mission
- Conducted flight tests using an aircraft simulator to evaluate performance and stability characteristics
- **Tools:** XFLR5, Plane Maker, X-Plane, MATLAB

Autonomous Delivery Using Quadcopter 2024

- Integrated **Pixhawk 4 and PX4** for autonomous waypoint navigation and mission execution
- Designed and implemented a servo-actuated payload release mechanism triggered through QGroundControl mission commands
- Conducted autonomous flight testing and analyzed flight logs, actuator response, trajectory, and velocity profiles
- **Tools:** PX4, QGroundControl

Design and Fabrication of Small-Scale Wooden Aircraft Model 2024

- Designed an aircraft model in CATIA considering aerodynamic and structural requirements
- Performed aerodynamic analysis in XFLR5 to evaluate lift, drag, and stability characteristics
- Fabricated a wooden prototype based on the CAD model

Airfoil Optimization Using Genetic Algorithm, SIC Hackathon 2024

- Optimized airfoil performance using genetic algorithms integrated with XFOIL
- Combined aerodynamic analysis, programming, and optimization techniques
- **Tools:** Python, XFOIL

Professional and Technical Activities

Member, Flying Club, Pulchowk Campus 2024–Present

- Contributed to UAV fabrication, system integration, and flight testing activities
- Collaborated with aerospace engineering teams on hands-on UAV design and development projects
- Conducted autonomous flight operations and evaluated UAV flight performance

Event Coordinator, National Mechanical and Aerospace Exhibition, Pulchowk Campus 2025

- Coordinated engineering-student event (RC car race) and technical workshops
- Organized technical competitions and the MechTRIX exhibition

Volunteer, National Mechanical and Aerospace Exhibition, Pulchowk Campus 2023

- Volunteered different Events such as Robo-soccer and Go-kart race.

Courses and Training

ATR-72 Aircraft Performance Weight and Balance Training, Yeti Airlines Ltd.	Oct 2025
Training on Dangerous Goods Regulations, Yeti Airlines Ltd.	Oct 2025
Training on Human Factor in Aviation, Yeti Airlines Ltd.	Oct 2025
Training on Aviation Security and Safety Management System (SMS), Yeti Airlines Ltd.	Oct 2025
Coding and Programming, Samsung Innovation Campus	2024

Key Competencies

Programming Languages: C, MATLAB, Python

Applications: CATIA, ANSYS, RDWorks, X-Plane, XFLR5, QGIS, QGC, MS Office, MS Excel

References

Kamal Darlami, Assistant Professor, Former Deputy Head of Department

Pulchowk Campus, Institute of Engineering, Tribhuvan University

Currently Pursuing PhD in University of Southern Queensland